COMP 441 | LABORATORY REPORT

Lab 7: Mutation Testing & Suite Effectiveness

AI-generated, manual, and combined test-suite evaluation

Prepared: 25 October 2026 **Project:** Task Manager sample repository
Lab 6 targets: `load_tasks`, `save_tasks`, and `find_task_by_title` in `app/tasks.py`

Outcome: initial suites killed 23 of 30 eligible mutants (76.7%). Two focused tests were added to verify explicit file modes and UTF-8 encoding; the updated combined suite killed all 30 eligible mutants (100%).

1. Method and scope

Mutation testing was run with mutmut 3.8.0 under Python 3.14.4 in WSL (mutmut does not support native Windows). Each run targeted `app/tasks.py` and selected the AI-only, manual-only, or combined Lab 6 tests. Covered-line mutation was enabled. The AI suite contains 7 tests, the manual suite 6 tests, and the combined suite 13 tests.

mutmut generated 31 candidates per run. One `add_task` candidate was classified as having no tests; it is reported separately and excluded from the mutation-score denominator. Score = killed / (killed + survived), giving 30 eligible mutants.

2. Mutation reports: original Lab 6 suites

Suite	Tests	Generated mutants	Eligible	Killed	Survived	No tests	Mutation score
AI-only	7	31	30	23	7	1	**76.7%**
Manual-only	6	31	30	23	7	1	**76.7%**
Combined	13	31	30	23	7	1	**76.7%**

The equal scores show that the two original suites had substantial overlap in what they asserted: combining them did not kill additional mutants. Seven equivalent or behaviorally indistinguishable changes survived in each original run.

Three surviving mutants analyzed

Mutant / change	Why the original assertions did not kill it
`load_tasks__mutmut_4` `encoding="utf-8" -> None`	The tests checked loaded JSON data from ASCII fixtures. In the UTF-8 test environment the default encoding produced the same result, so content-only assertions passed.
`load_tasks__mutmut_6` Explicit `"r"` mode removed	Omitting the mode makes Python use its default read mode. Existing assertions verified the returned list, not the mode passed to open, so behavior was unchanged.
`save_tasks__mutmut_3` `encoding="utf-8" -> None`	The save tests checked serialized JSON values. The platform default encoding produced identical output; no assertion protected the explicit UTF-8 portability contract.

The other four baseline survivors made similar changes to explicit encoding arguments, including removing the encoding or changing its capitalization.

3. Targeted tests and improved mutation report

Two regression tests were added in `tests/test_lab7_survivors.py`. They wrap the real built-in `open`, record its arguments, and still perform the actual file operation:

New test	Assertions and mutants addressed
`test_load_tasks_uses_explicit_read_mode_and_utf8`	Asserts the file is opened with mode `r` and `encoding="utf-8"`. Kills load-mode and encoding survivors, including the three analyzed variants.
`test_save_tasks_uses_explicit_write_mode_and_utf8`	Asserts the file is opened with mode `w` and `encoding="utf-8"`. Kills corresponding save-encoding survivors.

Final combined mutation run

Suite	Tests	Generated mutants	Eligible	Killed	Survived	No tests	Score
AI + manual + Lab 7 tests	15	31	30	30	0	1	**100%**

The updated combined suite killed all seven previously surviving mutants. The one no-tests candidate remained excluded, so the final eligible score improved from 23/30 (76.7%) to 30/30 (100%).

Validation: `python -m pytest -q tests` completed with **21 passed**. This includes 13 Lab 6 tests, two new targeted tests, and six existing core task tests.

4. Coverage comparison and conclusion

The submitted Lab 6 coverage report records 48% statement coverage for AI-only, manual-only, and combined, measured across all of `app/tasks.py`. The original mutation scores were 76.7%. These percentages are not interchangeable: coverage counts executed statements across the module, while mutation testing measures whether assertions detect changes in selected, covered code.

The gap shows why coverage alone is insufficient: lines can execute without assertions protecting important contracts. Here, content-level assertions let explicit file-mode and encoding changes survive. Focused argument assertions raised the final selected-scope mutation score to 100%. This does not establish full quality for every function in the module; one candidate had no tests, and the experiment is scoped to the Lab 6 targets.

Conclusion: the original suites exercised relevant code but left file-I/O guarantees unasserted. Two narrowly targeted tests killed all seven survivors and measurably improved the final combined mutation score.

Reproducibility: mutation runs used mutmut 3.8.0 with source `app/tasks.py`, covered-line mutation enabled, and AI-only, manual-only, and combined test selections. The updated root `setup.cfg` selects the complete `tests` directory for the final run.